

Timothy Cai

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[portfolio](#) | [linkedin](#)

EDUCATION

Bachelor of Engineering - Mechanical (Mechatronics)

University of British Columbia | 3.7/4.0 GPA

Sep 2022 - May 2027

Vancouver, BC

SKILLS

Mechanical SOLIDWORKS | OnShape | NX | FEA & CFD (Fluids + Thermal) | GD&T | Fluid & Vacuum Systems

Electronics & Software KiCAD | EAGLE | Sensors & DAQ | C++ | Python | Control Algorithms | Test Automation

Manufacturing 400+ hrs Mill, Lathe, Waterjet | DFMA for FDM, SLA and DMLS | CAM | PCB Assembly

PROJECT HIGHLIGHTS (SEE PORTFOLIO)

- Currently engineering and testing a bipropellant rocket with a **regeneratively cooled**, 3.5kN ethanol/nitrous rocket engine with a **paired like-doublet** injector and **CFD-driven** geometry.
- Designed and flew a self-stabilized thrust vectoring rocket, with **guidance electronics** designed in EAGLE and control algorithms written in **C++ and Python**, achieving **±3.5 degrees of stability**.
- Created and analyzed a **supersonic rocket** in **SOLIDWORKS**, with the rocket reaching a top speed of **1280km/hr** and an altitude of **2.8km** with successful recovery.

EXPERIENCE

Rocket Lab

May 2026 - Aug 2026

Development Intern

Auckland, NZ

- Designed additively manufactured injector elements for a classified in-space propulsion system, with accompanying analysis software and quality control workflows.
- Brought up and overhauled a fluid test rig, from calibrating Coriolis flowmeters and loadcells to automating discharge coefficient and Reynolds number calculation across thousands of test cycles.

Reflect Orbital

Jan 2026 - Apr 2026

Mechanical Intern

Hawthorne, CA

- **Resolved mission critical deployment failure** by designing a **>67N-m stabilizing** boundary condition device with an aramid lockdown mechanism, within a frozen design space.
- Designed, qualified, and integrated compliant **hold-down release mechanisms** for the main payload inline with aerospace standards and practices, including full component release flows.
- Built a **qualification vehicle** payload to specification for **vibration testing** in conjunction with writing **manufacturing flow process** documents.

Atomic Semi

May 2025 - Aug 2025

Mechanical Intern

San Francisco, CA

- Invented a **solid state "tractor beam"** for wafer transport by exploiting squeeze-film levitation and local Bernoulli effects through a vacuum feedthrough Langevin transducer.
- Optimized the gripper design with **modal analysis**, holding objects stably in three axes at a 40µm distance with under 5W of steady state power draw.
- Engineered an **automated butterfly vacuum valve**, achieving better-than-industry **helium-tight** sealing at 10⁻¹⁰ torr-l/s while both reducing manufacturing **costs by 86%** and **extending lifetime** by over 15x.
- Rapidly prototyped and installed an **automated ALD vacuum chamber door** that was productionized and deployed, eliminating manual operation and supporting **thousands** of lights-out operational cycles.

ARTMS Inc

May 2024 - Dec 2024

Mechanical/Mechatronics Intern

Burnaby, BC

- Designed and implemented a solid target **leak check station** utilizing Festo and Swagelok pneumatic components, resulting in a **increase to 100%** target manufacturing success.
- Generated schematics and board layouts with constant current and voltage architecture for an **automated radioisotope recycling** product using KiCAD, reducing space by 85%.

DI Self-Composites

Aug 2023 - Sept 2024

Mechatronics Engineering Intern

Vancouver, BC

- Utilized **SOLIDWORKS** to rapidly design and prototype a movement system capable of **printing metal** at 400mm/min in a highly volatile and **abrasive environment 20% under budget**.
- Adapted existing Marlin code and **wrote custom firmware** to quickly produce a **full-featured** printing solution.